might then be inappropriate. The model might be valid for today's climate, but not extrapolatable to others.

One way to test this program is to apply it to the very different climates of other planets. Can it predict the structure of the atmosphere on Mars and the climate there? The weather? What about Venus? If it were to fail these test cases, we would be right in mistrusting it when it makes predictions for our own planet. In fact, climate models now in use do very well in predicting from first principles of physics the climates on Venus and Mars.

On Earth, huge upwellings of molten lava are known and attributed to superplumes convecting up from the deep mantle and generating vast plateaus of frozen basalt. A spectacular example occurred about a hundred million years ago, and added perhaps ten times the present carbon dioxide content to the atmosphere, inducing substantial global warming. These plumes, it is thought, occur episodically throughout Earth's history. Similar mantle upwelling seem to have occurred on Mars and Venus. There are sound practical reasons for us to want to understand how a major change to the Earth's surface and climate could suddenly arrive unannounced from hundreds of kilometers beneath our feet.

Some of the most important recent work on global warming has been done by James Hansen and his colleagues at the Goddard Institute for Space Sciences, a NASA facility in New York City. Hansen developed one of the major computer climate models and employed it to predict what will happen to our climate as the greenhouse gases continue to build up. He has been in the forefront of testing these models against ancient climates of the Earth. (During the last ice ages, it is of interest to note, more carbon dioxide and methane are strikingly correlated with higher temperatures.) Hansen collected a wide range of weather data from this century and last, to see what actually happened to the global temperature, and then compared it to the computer model's predictions of what *should* have happened. The two agree to within the errors of measurement and calculation, respectively. He courageously testified before Congress in the face of a politically generated order from the White House Office of Management and